

VAIBHAV PATEL

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PROFESSIONAL SUMMARY

Full Stack Engineer with 3+ years of experience building and operating scalable distributed systems in production environments. Specializes in AWS serverless architectures (Lambda, AppSync, DynamoDB) and owning complex features end-to-end. Proven track record of shipping high-impact features: designing resilient systems for mission-critical services, optimizing performance through intelligent caching, and proactively catching edge cases before they affect users. Experienced in production troubleshooting using observability tools (Dynatrace, Splunk), cross-timezone team coordination, and driving architectural decisions. Proficient in Node.js, Go, Python, JavaScript, and modern DevOps practices.

TECHNICAL SKILLS

Languages: JavaScript, Python, Go, Java, SQL, TypeScript

Cloud & Databases: Firebase (Cloud Functions, Firestore, Hosting), AWS (Lambda, AppSync, DynamoDB), PostgreSQL, Redis, MongoDB, MySQL

Frameworks: Node.js/Express, React, Next.js, Laravel, Flask, React Native

AI & Tools: OpenAI API, Embeddings, Dynatrace, Splunk, GitHub Actions, Docker, Terraform, Git

EXPERIENCE

Backend Developer

Jun. 2025 – Present

Air Canada (via TCS)

Remote

- Architected and owned the **Disruption Resolution Engine**, handling flight disruptions (cancellations, cabin changes, time/flight changes, operational delays) for itinerary reconciliation in mission-critical systems
- Redesigned system by separating regular and disruption flows, eliminating circular dependencies and enabling rapid feature addition while **currently handling 50% of production traffic**
- Engineered **BFS-based graph traversal algorithm** to correctly match disrupted segments across complex PNR types (one-way, round-trip, multi-city). Proactively caught critical edge case in operational delay handling that would have silently skipped segments from user responses
- Optimized performance through intelligent **DynamoDB caching**, eliminating redundant EIP calls and expensive graph calculations for repeated PNRs
- Coordinated cross-team integration of meal preorder feature across timezones, implementing graceful error handling for external API failures and driving operational excellence through proactive observability using Dynatrace and Splunk

Full Stack Developer

Mar. 2024 – Jun. 2025

Willekk Digital Solutions

Remote

Project 1: PowHerFit

- Developed robust web applications using **Laravel** framework, leveraging **MVC** architecture, eloquent **ORM**, and blade templating engine to deliver efficient and scalable solutions
- Implemented staging environment on **AWS**, reducing production deployment risks by enabling thorough API and CMS testing before code promotion
- Configured **Nginx** servers on **EC2** instances, resulting in optimized performance and enhanced communication between backend services and frontend interfaces, improving overall application responsiveness
- Established CI/CD pipelines using **Github Actions**, automating the build, test, and deployment processes, resulting in a **40%** reduction in deployment time and a decrease in production incidents

Project 2: RealOnes

- Collaborated with cross-functional teams to swiftly address and rectify **35** reported issues using Android Studio, demonstrating strong problem-solving skills and attention to detail
- Identified performance bottlenecks in API responses generated from **AWS Lambda** functions, leading to **10%** optimization improvements
- Resolved critical issues causing disruptions in the business dashboard functionality, ensuring seamless operation for users and enhancing overall user experience

Cloud Engineer Co-op

Sep. 2023 – Dec. 2023

Manulife

Toronto, ON

- Achieved a significant reduction in subscription expenses by automating the deletion of expired and unused resources through the development of **Python**-based automation script, utilizing **Azure Automation**, **Azure Resource Graph**

- Automated provisioning of Azure resources through **Terraform**, reducing deployment time by **10%** and increasing scalability and reliability for clients
- Contributed to the improvement of code quality through active participation in code reviews, ensuring adherence to best practices and enhancing overall team efficiency

Software Developer (Part-Time)

Jan. 2023 – Aug. 2023

Gravioty Research

Remote

- Developed a multi-currency cryptocurrency wallet from scratch, providing support for Ethereum and Bitcoin blockchains using **React**, **Web3.js**, **Ethers.js**, and **Bitcoinjs-lib**
- Designed and implemented a robust state management architecture for the wallet utilizing **Redux**, resulting in optimized data handling
- Integrated **WebSocket** for real-time updates on cryptocurrency prices, ensuring users have access to the most up-to-date information
- Increased cross-browser compatibility by devising dynamic and easily configurable **webpack** scripts, ensuring seamless user experience across different browsers

PROJECTS

Insights | *React 19, Node.js/Express, TypeScript, PostgreSQL, Redis, Firebase, Vite*



- Built full-stack financial intelligence platform enabling users to track multi-account banking, manage investment portfolios, create budgets, and receive AI-powered financial recommendations. Architected with **React 19 + Redux Toolkit** frontend and **Firebase Cloud Functions** backend
- Implemented sophisticated **AI transaction categorization** engine using **OpenAI embeddings** for semantic merchant classification, with **user preference override** support. Built **streaming chat interface** with **OpenAI API** for interactive financial queries, budget creation, and investment analysis
- Designed intelligent caching strategy with **Redis** and **Upstash**, reducing API response times by **50%**. Implemented embedding cache for merchant lookups and user category preferences to eliminate redundant OpenAI calls
- Engineered comprehensive test suite with **160+ Jest tests** covering financial calculations, category classification, and budget logic. Integrated **CI/CD pipelines** and automated email parsing for seamless transaction import
- Designed rich interactive dashboard with real-time net worth tracking, multi-asset allocation visualization, cash flow analysis, and goal progress monitoring using **Tailwind CSS** and **DaisyUI**

OGSA | *Firebase, React, JavaScript, Stripe, GCP*



- Implemented role-based access control, enabling seamless management of permissions, resulting in enhanced security and efficient utilization of features by **500+ users**
- Developed a robust ticketing system allowing for the creation and management of event tickets, including early bird options, leading to a **5%** increase in ticket sales and revenue generation
- Utilized **Stripe API** and **Firebase Cloud Functions** for seamless payment processing, resulting in secure transactions and streamlined revenue management, with over **800+** successful payments processed

LEADERSHIP & COMMUNITY INVOLVEMENT

Founder

Jun. 2021 – Present

Ontario Gujarati Students Association

Toronto, ON

- Lead an independent cultural group, organizing, and hosting over **5 events** annually with more than **700 attendees**
- Established a strong online presence through social media platforms, resulting in a **30%** increase in event attendance and engagement
- Coordinated aspects such as venue selection, budgeting, and team coordination, to ensure flawless execution
- Managed a team of **10+** individuals, delegating tasks and responsibilities to ensure the smooth running of events

EDUCATION

Wilfrid Laurier University

Waterloo, ON

Honours Bachelor of Science in Computer Science

Sep. 2019 – Dec. 2023

- Awarded Dean's List (2020 - 2021)

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals

Mar. 2024

Microsoft